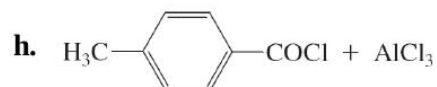
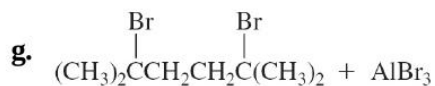
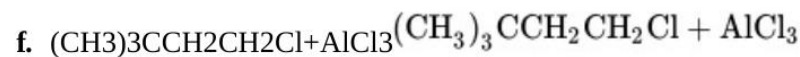
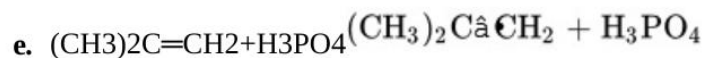
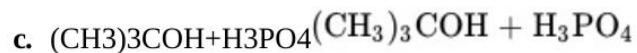
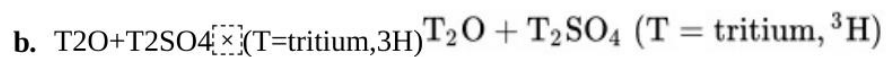


48. Give the expected major product of addition of each of the following reagent mixtures to benzene. (**Hint:** Look for analogies to the reactions presented in this chapter.)



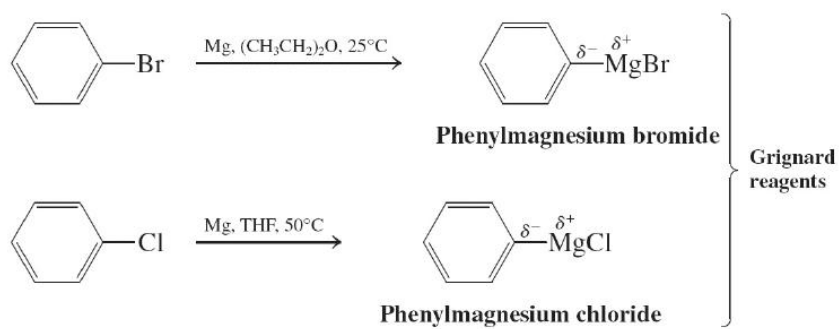
49. Write mechanisms for the reactions of parts (c) and (f) in [Problem 48](#).

50. Hexadeuteriobenzene, C_6D_6 , is a very useful solvent for ^1H NMR spectroscopy because it dissolves a wide variety of organic compounds and, being aromatic, is very stable. Suggest a method for the preparation of C_6D_6 .

53. (a) 3-Phenylpropanoyl chloride, $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{COCl}$, reacts with AlCl_3 to give a single product with the formula $\text{C}_9\text{H}_8\text{O}$ and an ^1H NMR spectrum with signals at $\delta = 2.53$ (t, $J = 8$ Hz, 2 H), $\delta = 3.02$ (t, $J = 8$ Hz, 2 H), and $\delta = 7.2-7.7$ (m, 4 H) ppm. Propose a structure and a mechanism for the formation of this derivative.
- (b) The product of the process described in (a) is subjected to the following reaction sequence: (1) NaBH_4 , $\text{CH}_3\text{CH}_2\text{OH}$; (2) conc. H_2SO_4 , 100°C ; (3) H_2 , $\text{Pd}-\text{C}$, $\text{CH}_3\text{CH}_2\text{OH}$. The resulting molecule exhibits five resonance lines in its ^{13}C NMR spectrum. What is the structure of the substance formed after each of the steps in this sequence?

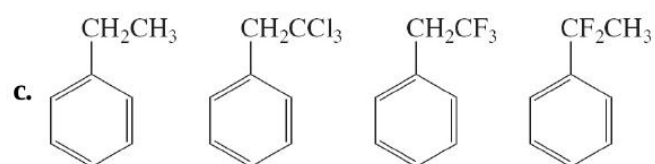
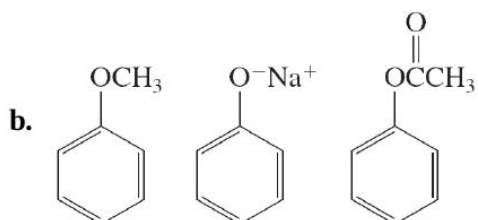
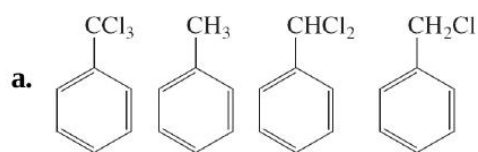
54. The text states that alkylated benzenes are more susceptible to electrophilic attack than is benzene itself. Draw a graph like that in [Figure 15-20](#) to show how the energy profile of electrophilic substitution of methylbenzene (toluene) would differ quantitatively from that of benzene.

55. Like haloalkanes, haloarenes are readily converted into organometallic reagents, which are sources of nucleophilic carbon.

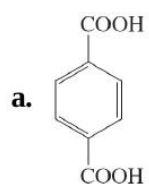


56. Give efficient syntheses of the following molecules, beginning with benzene. **(a)** 1-Phenyl-1-heptanol; **(b)** 2-phenyl-2-butanol; **(c)** octylbenzene. **(Hint:** Use a method from [Section 15-13](#). Why will Friedel-Crafts alkylation not work?)

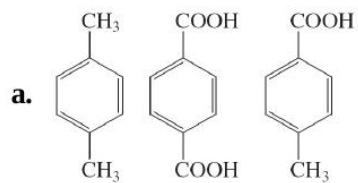
30. Rank the compounds in each of the following groups in order of decreasing reactivity toward electrophilic substitution. Explain your rankings.



32. Specify whether you expect the benzene rings in the following molecules to be activated or deactivated.



33. Rank the compounds in each of the following groups in order of decreasing reactivity toward electrophilic aromatic substitution. Explain your answers.



34. Halogenation of 1,3-dimethylbenzene (*m*-xylene) takes place 100 times faster than halogenation of either its 1,2- or 1,4-isomers (*o*- or *p*-xylene). Suggest an explanation.

35. Write the structure(s) of the major product(s) that you expect from each of the following electrophilic aromatic substitutions. **(a)** Nitration of methylbenzene (toluene); **(b)** sulfonation of methylbenzene (toluene); **(c)** nitration of 1,1-dimethylethylbenzene (*tert*-butylbenzene); **(d)** sulfonation of 1,1-dimethylethylbenzene (*tert*-butylbenzene).

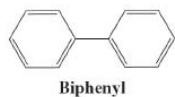
In what way does changing the substrate structures from methylbenzene (toluene) in (a) and (b) to 1,1-dimethylethylbenzene (*tert*-butylbenzene) in (c) and (d) affect the expected product distributions?

36. Write the structure(s) of the major product(s) that you expect from each of the following electrophilic aromatic substitutions. **(a)** Sulfonation of methoxybenzene (anisole); **(b)** bromination of nitrobenzene; **(c)** nitration of benzoic acid; **(d)** Friedel-Crafts acetylation of chlorobenzene.

37. Draw appropriate resonance forms to explain the deactivating meta-directing character of the SO_3H group in benzenesulfonic acid.

- 38.** Do you agree with the following statement? “Strongly electron-withdrawing substituents on benzene rings are meta directing because they deactivate the meta positions less than they deactivate the ortho and para positions.” Explain your answer.

- 39.** Draw appropriate resonance forms to explain the activating ortho, para-directing character of the phenyl substituent in biphenyl (below).



40. Give the expected major product(s) of each of the following